PART I: ARTIFICIAL NEURAL NETWORK

HISTORY

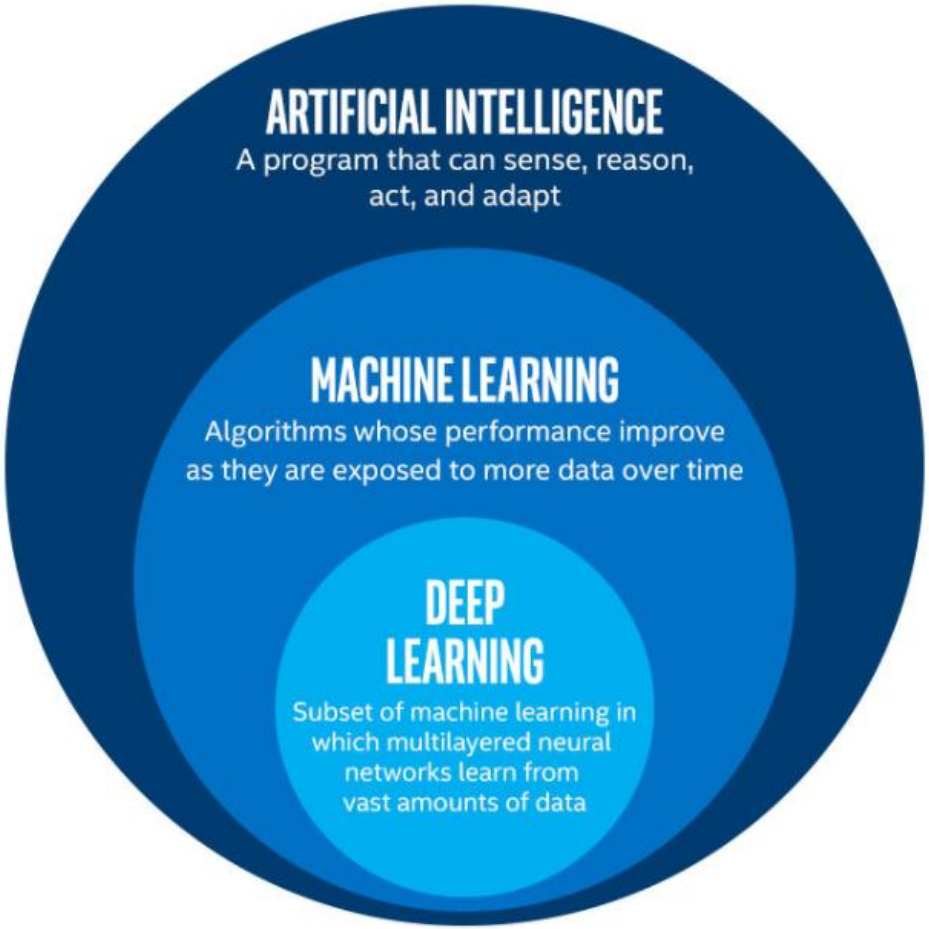
- People started investigating special mechanisms that act as the basis of human intelligence.
- They managed to create machines that imitate part of these mechanisms.
- There are difficulties in imitating all such processes as they **mostly rely on human interactions.**
- **ANN –** Simulate human thinking

EXAMPLES

1. Imitating the mechanisms needed to win a chess match
2. Imitating the mechanism to translate text into a different language in accordance with its grammatical rules.

AI or machine intelligence is a field of study that aims to give **cognitive powers** to computers to program them to learn and solve problems.

Machine Learning & Deep Learning



MCQ 1

What's the main point of difference between human & machine intelligence?

(i) Human have emotions and machines do not.

(ii) Human have more IQ & intellect that the machines.

(iii) Human perceive everything as a pattern while machine perceive it merely as data.

(iv) Human have sense organs unlike machines.

(v) None of the above

Artificial Neural Network Algorithm

State of the art Technique for many **machine learning algorithms**

Why do we need ANN algorithm?

To solve **complex, nonlinear hypothesis**

MCQ 2

The advantage/s of ANN is/are,

(A). Ability to learn how to do tasks based on the data given for training or initial experience.

(B).Creating its own organization or representation of the information it receives during learning time.

(C). Capacity to conduct parallel computing

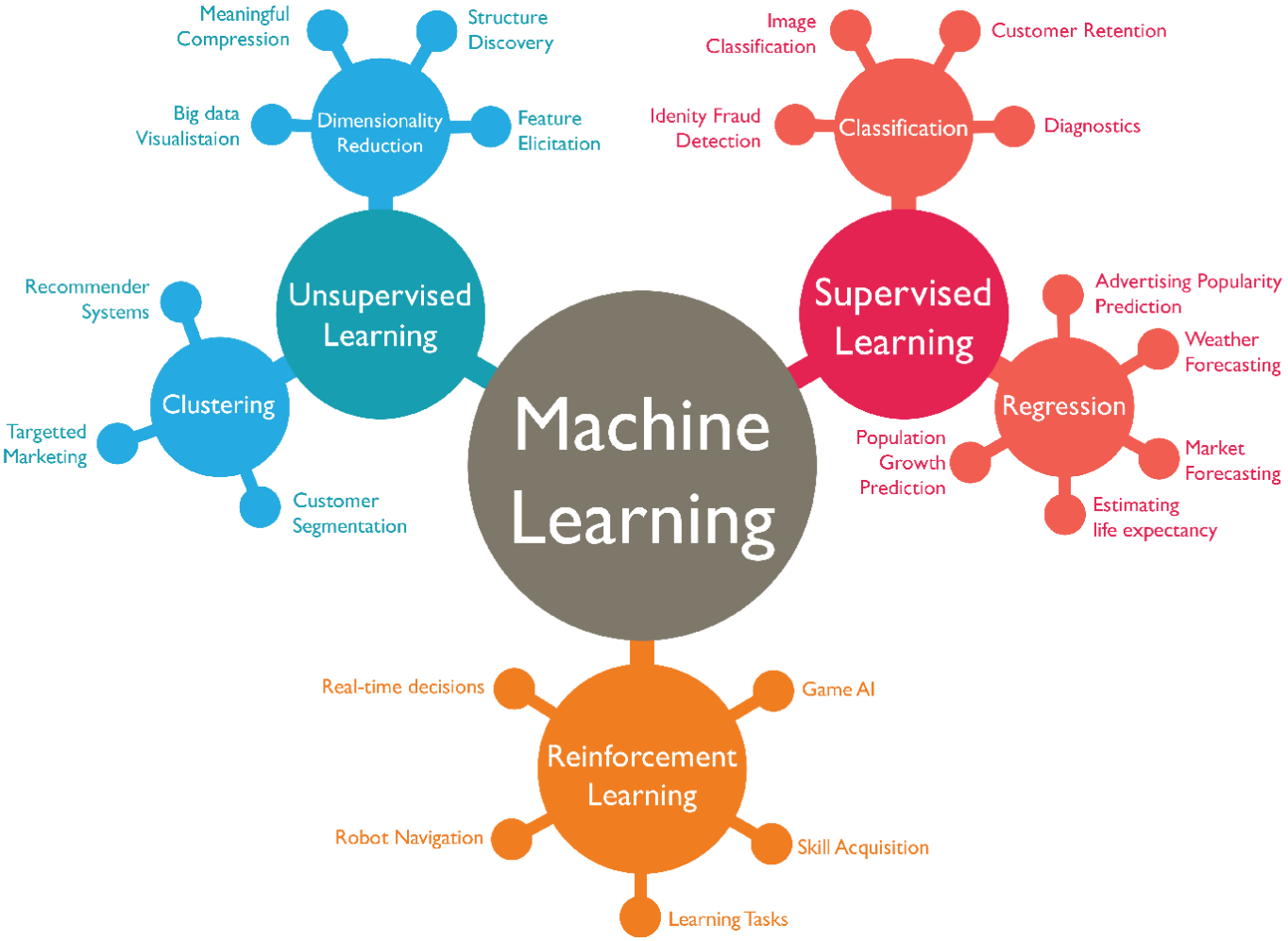
(i) (A) only.

(ii) (A) and (B) only.

(iii) (A) and (C) only.

(iv) (A), (B) and (C)

(v) None of (A), (B) and (C) is true.


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

MCQ 3

Which of the following is true about unsupervised learning?

(i) Features of group explicitly stated.

(ii) Number of groups may be known.

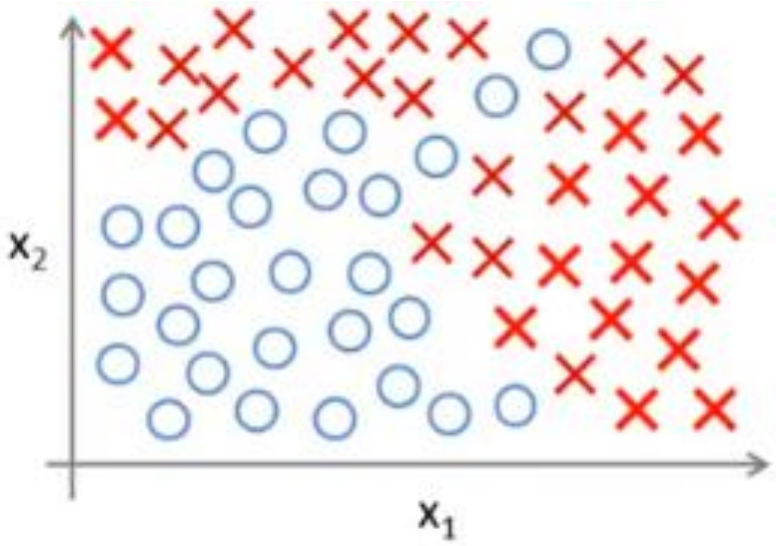
(iii) Neither feature nor number of groups is known.

(iv) There exist information about the targets.

(v) None of the above.

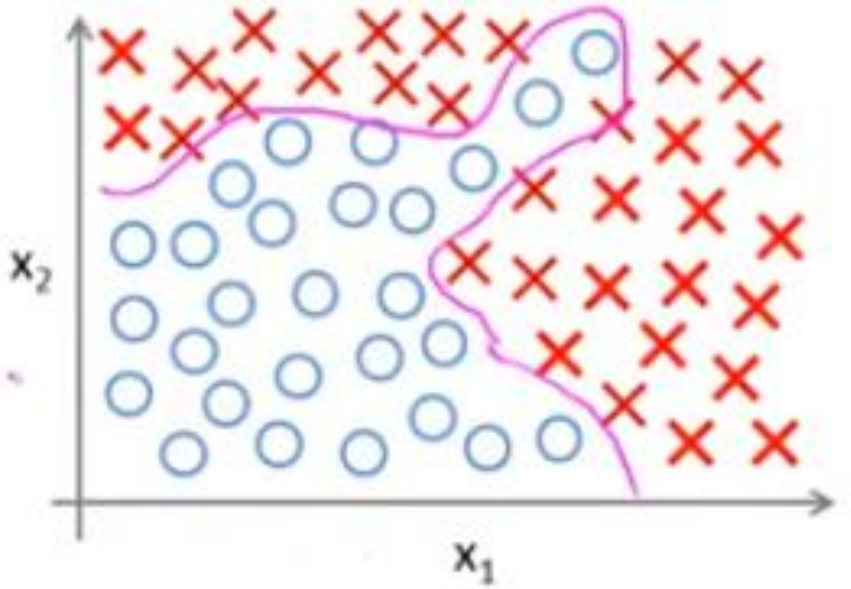
Nonlinear Hypothesis

Nonlinear Classification



Logistic Regression

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$



A Practical Application

Example: Housing classification problem.

$x_1 =$ size
$x_2 =$ # bedrooms
$x_3 =$ # floors
$x_4 =$ age
$\dots$
x_{100}

About 5000 features with only the second order polynomial and interactions!!!

LOT OF FEATURES!
OVERFITTING MIGHT OCCURE!
COMPUTATIONALLY EXPENSIVE!

Number of features are really high in most of the machine learning problems.

More Applications; Car Classification

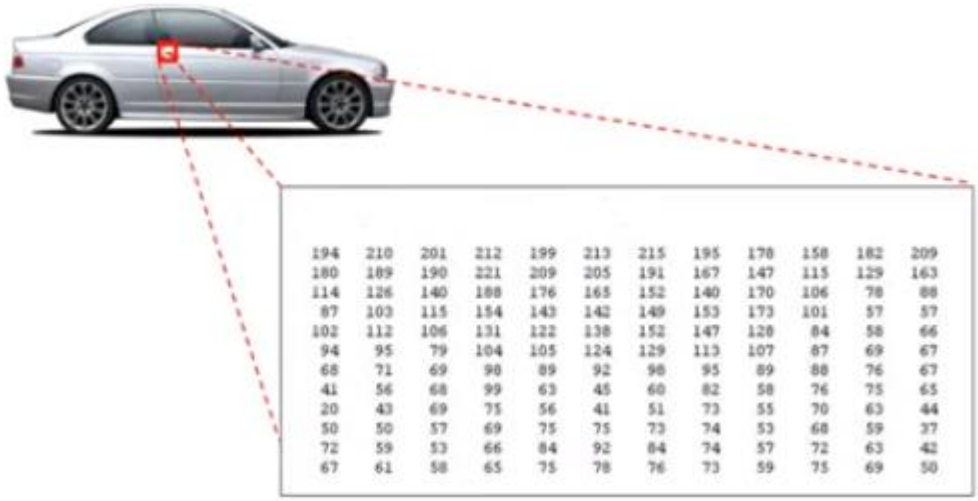
(Computer Vision)

Task: Train a classifier that examine a particular image and tells whether it is a car or not.

What you see…



What the camera see…



More Applications; Car Classification

(Computer Vision)

Car Detection Problem:

1. Look at the pixel intensity values
2. Tell that this matrix represents door handle of the car.

Method

- Labeled examples of cars
- Labeled examples of things that are not cars
- Divide the examples to train, test, validation
- Feed the training set to the neural network classifier

Illustration: Car Detection



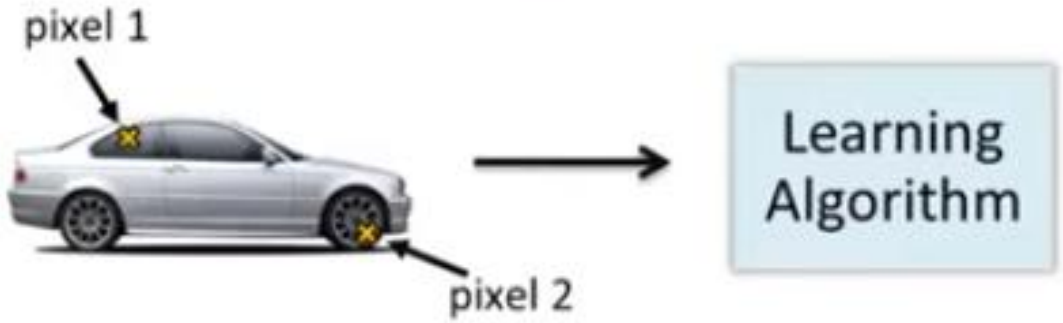


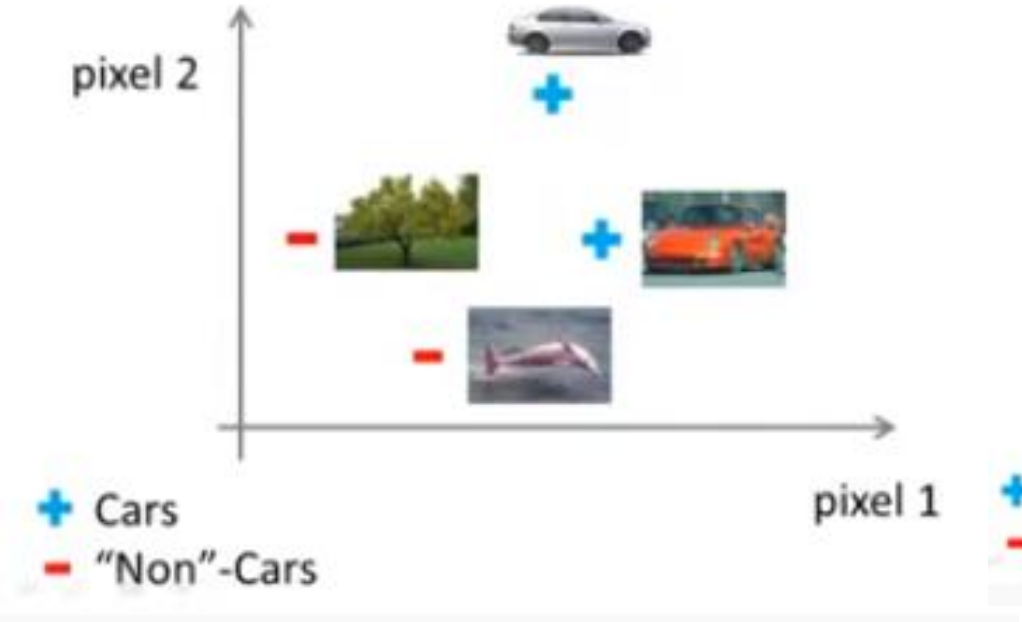
Testing:



What is this?

Illustration: Car Detection





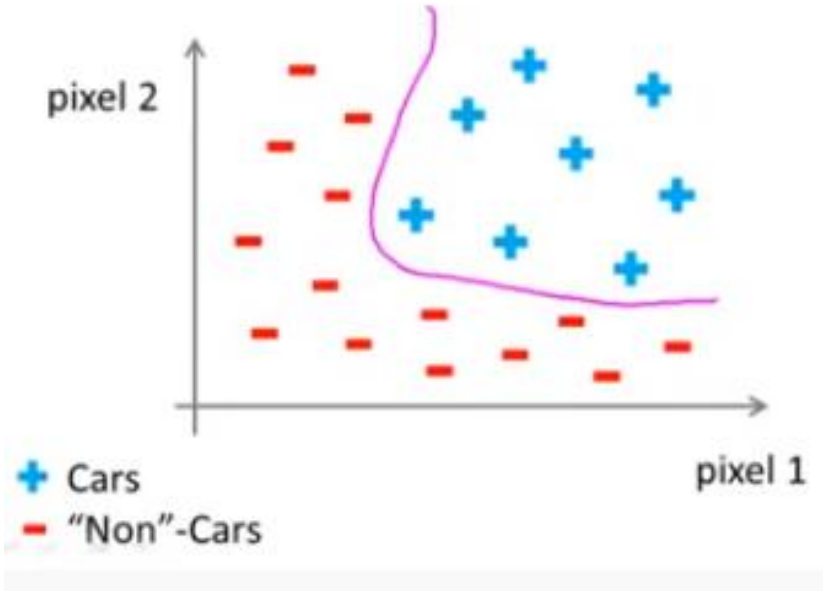


Illustration: Car Detection

50*50 pixel images >>> 2500 pixels

N = 2500 (7500 if RGB)

$$x = \begin{bmatrix} pixel\ 1\ intesnsity \\ pixel\ 2\ intensity \\ . \\ . \\ . \\ pixel\ 2500\ intensity \end{bmatrix}$$

All the quadratic features = 3 million features

Simple logistic regression with quadratic features is **not a good approach** to solve **complex nonlinear hypotheses**

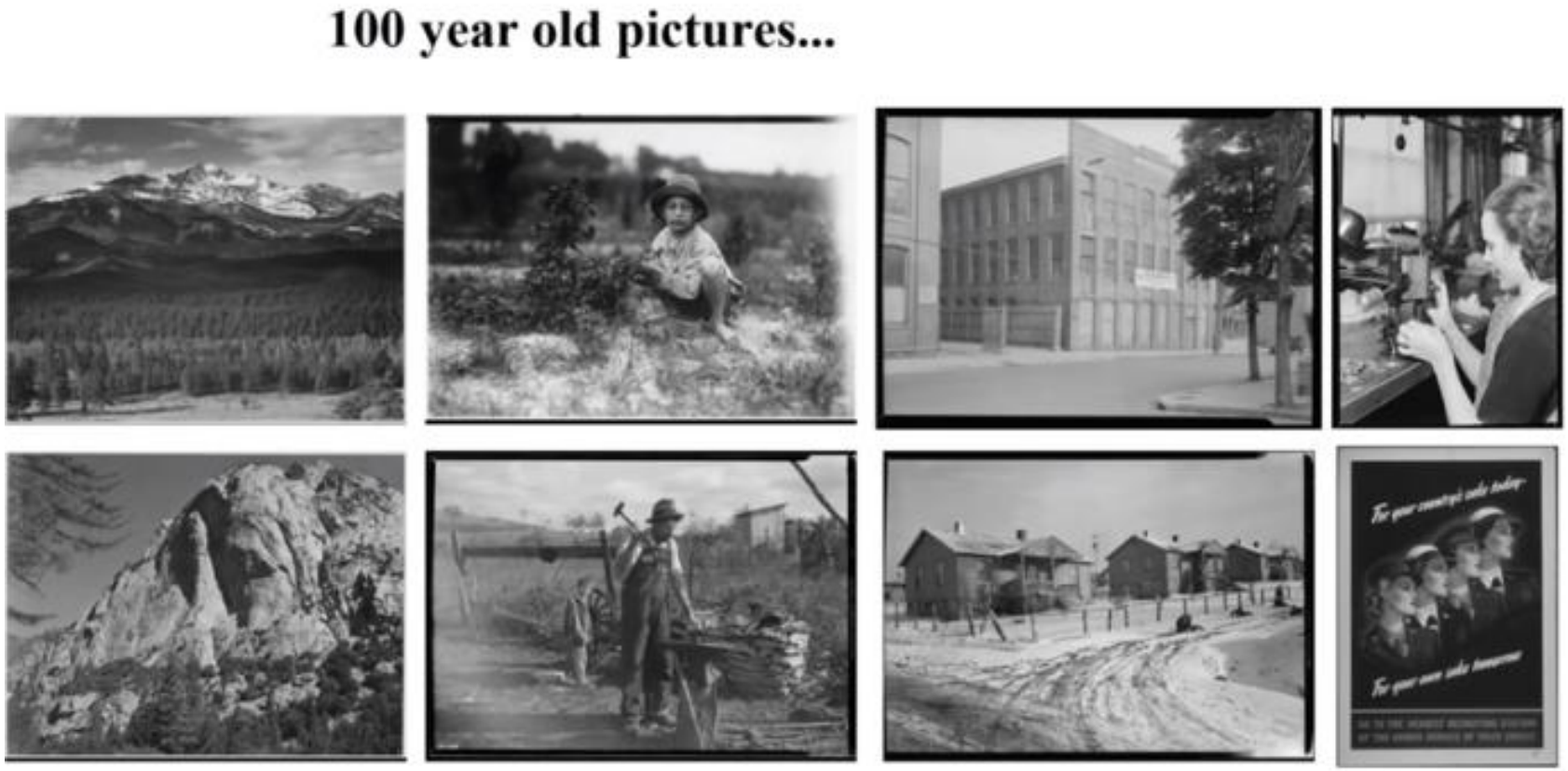
Top Applications

- Restore colours to black and white photos and videos.



Top Applications

- Restore colours to black and white photos and videos.



Top Applications

- Let there be color! – A convolution neural network classifier that automatically colorize grayscale images.











For your country's sake today
For your own sake tomorrow








For your country's sake today
For your own sake tomorrow

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Top Applications

- Pixel Enhancing – Deep neural network to take images with very low resolution and predict what those face will actually look like.



Top Applications

- Lip Reading – Can convert a silent video of a person talking into text.
- Average lip reader performance is only 52%
- LipNet achieves about 95% accuracy!!!

https://www.youtube.com/watch?v=fa5QGremQf8&feature=share

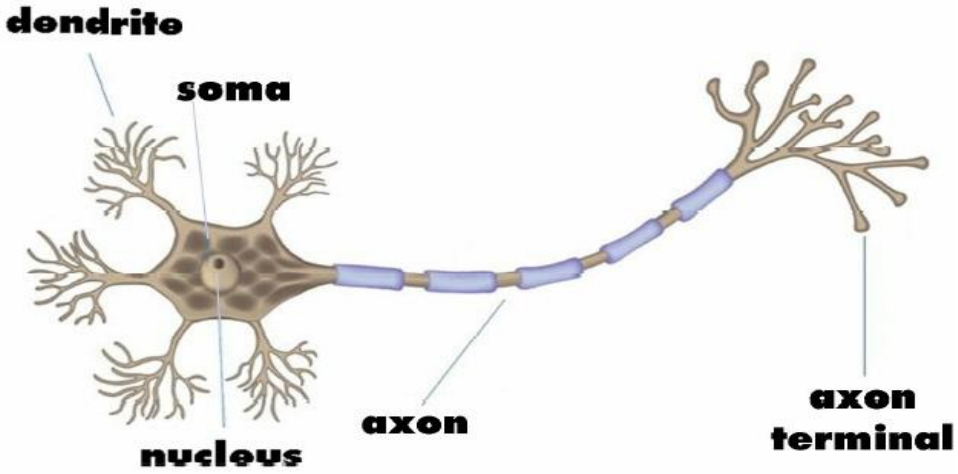
Advantages of ANN

- **Adaptive learning:** An ability to learn how to do tasks based on the data given for training or initial experience.

- **Self-Organisation:** An ANN can create its own organisation or representation of the information it receives during learning time.

- **Real Time Operation:** ANN computations may be carried out in parallel, and special hardware devices are being designed and manufactured which take advantage of this capability.

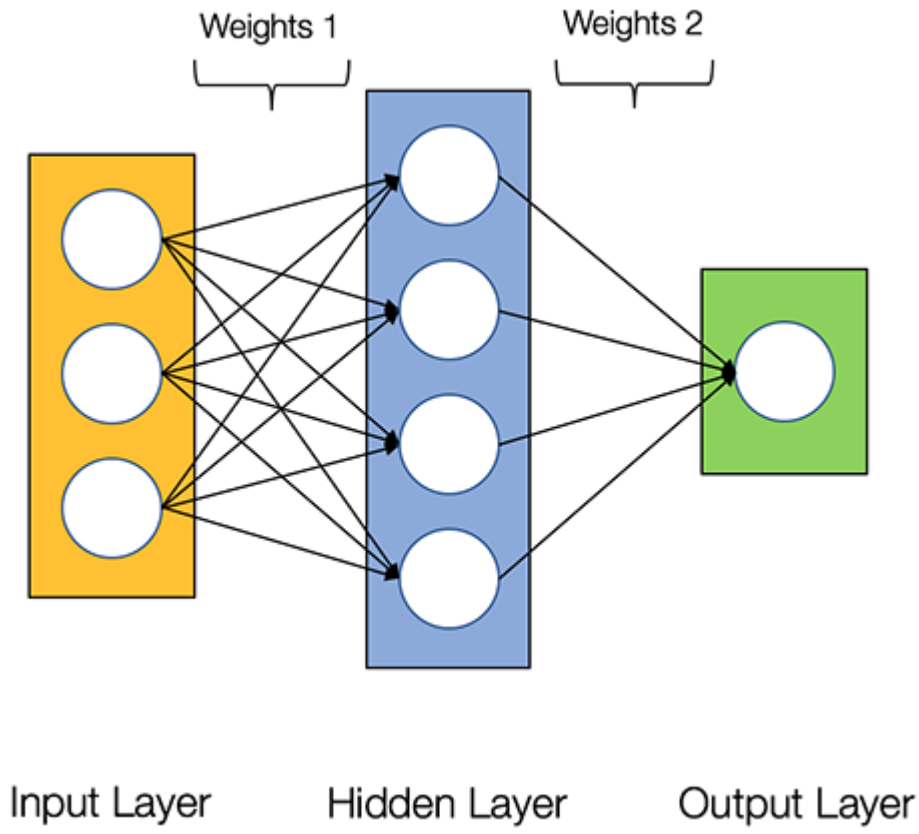
Inspiration of an Artificial Neural Network



- **Dendrites:** Entry points in each neuron which take input from other neurons in the network in form of electrical impulses
- **Cell Body:** It generates inferences from the dendrite inputs and decides what action to take
- **Axon terminals:** They transmit outputs in form of electrical impulses to next neuron

Structure of a Neural Network

ANNs define the **neuron** as a **central processing unit**



- Input layer
- Hidden layer
- Output layer
- Weights
- Bias
- Activation functions

Layered Architecture of ANN

- Any neural network processing a framework has a set **of inputs, a processor, and a set of outputs.**

- The inputs form the input layer, the middle layer(s) which performs the processing is called the hidden layer(s), and the output(s) forms the output layer.

MCQ 4

The number of nodes in the input layer is 10 and the hidden layer is 5. The maximum number of connections from the input layer to the hidden layer are,

i. 50.
ii. Less than 50.
iii. More than 50.
iv. In between 50 and 60.
v. It is an arbitrary value.

Weights

- A weight is multiplied to the input to add up to form the output.

- Weights are numerical parameters which determine how strongly each of the neurons affects the other.

- Output,

$$y = f(x) = \sum x_i w_i + b_i$$

Where i goes from 1 to the number of inputs considered

Matrix multiplication to arrive at the weighted sum

MCQ 5

An artificial neuron receives n inputs $x_1, x_2, x_3, \ldots, x_n$ with weights $w, w_2, w_3, \ldots, w_n$ attached to the input links. The weighted sum …… is computed to be passed on to a non-linear filter Φ called activation function to release the output.

(i) $\sum w_i$

(ii) $\sum x_i$

(iii) $\sum w_i + \sum x_i$

(iv) $\sum w_i * \sum x_i$

(v) None of the above

MCQ 6

A neuron with 3 inputs has the weight vector $[0.2 \ \ -0.1 \ \ 0.1]^T$ and a bias $\theta = 0$. If the input vector is $X = [0.2 \ \ 0.4 \ \ 0.2]^T$ then the total input to the neuron is:

(i) 0.20

(ii) 1.0

(iii) 0.02

(iv) -1.0

(v) None of the above.

Biases

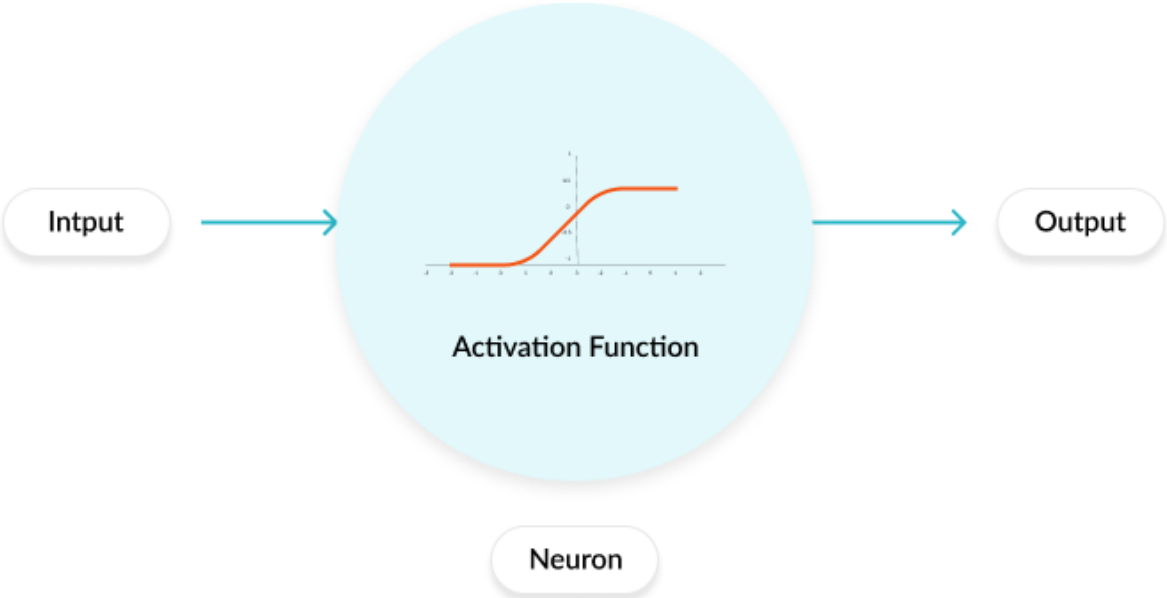
- Bias is like the intercept added in a linear equation.

- An additional parameter which is used to adjust the output along with the weighted sum of the inputs to the neuron.

$$output = sum(weights * inputs) + bias$$

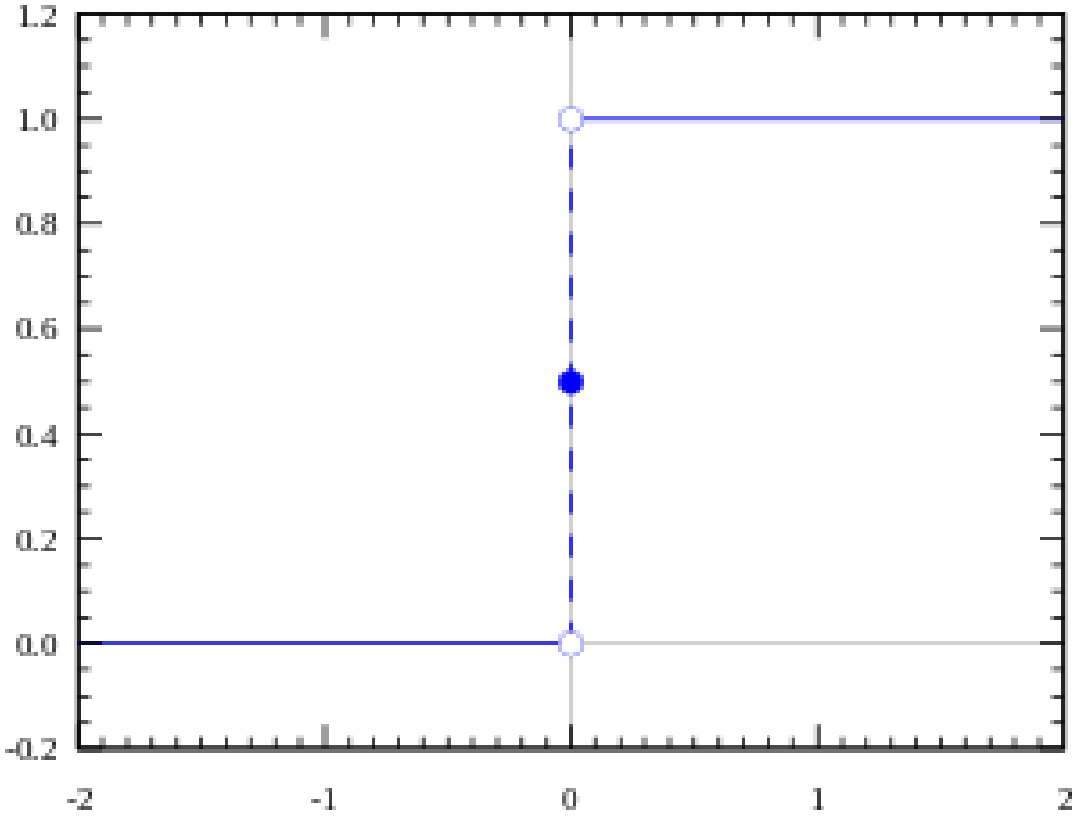
Activation Functions

- A mathematical function which converts the input to an output, and which adds nonlinearity to the model fitting.

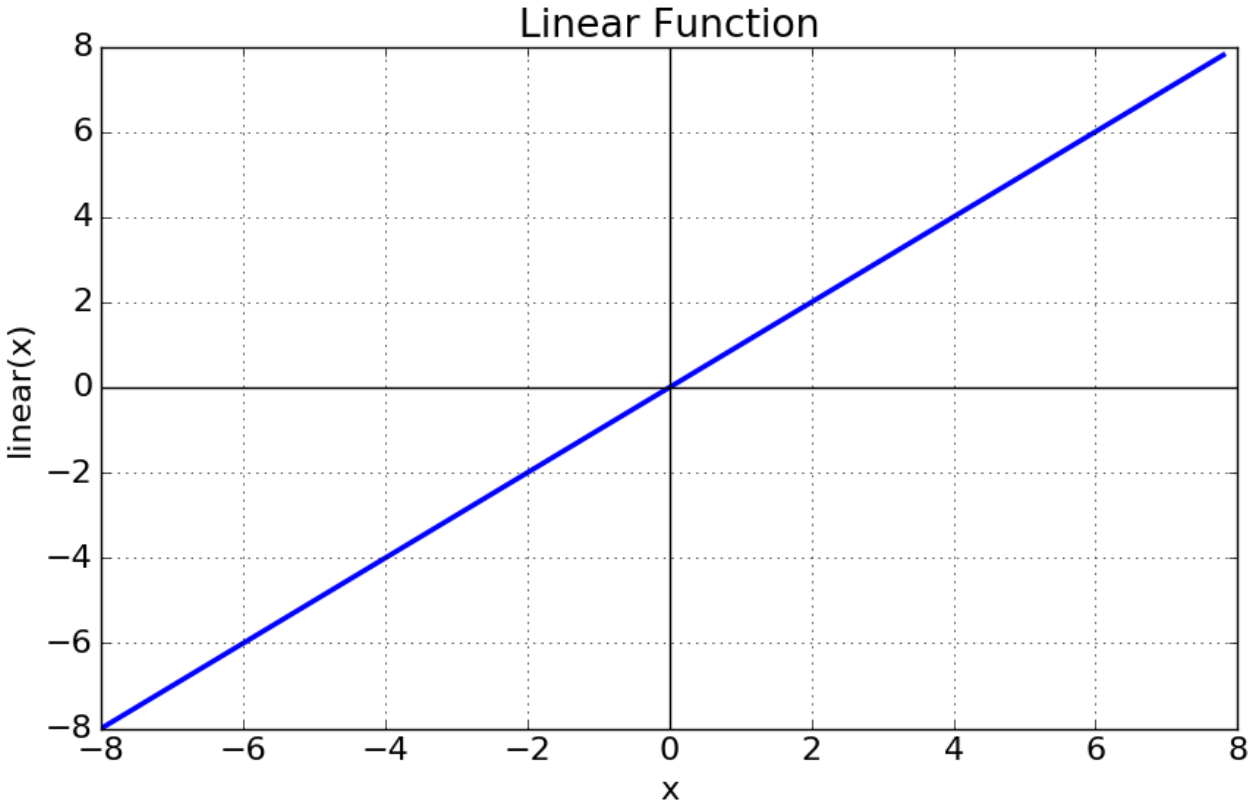


- Properties:
  - should be differential.
  - should be simple and fast in processing.

Binary Step Function



Linear Activation Function



Issues with Linear Step Function

1. Not possible to use backpropagation (gradient descent) to train the model

2. All layers of the neural network collapse into one

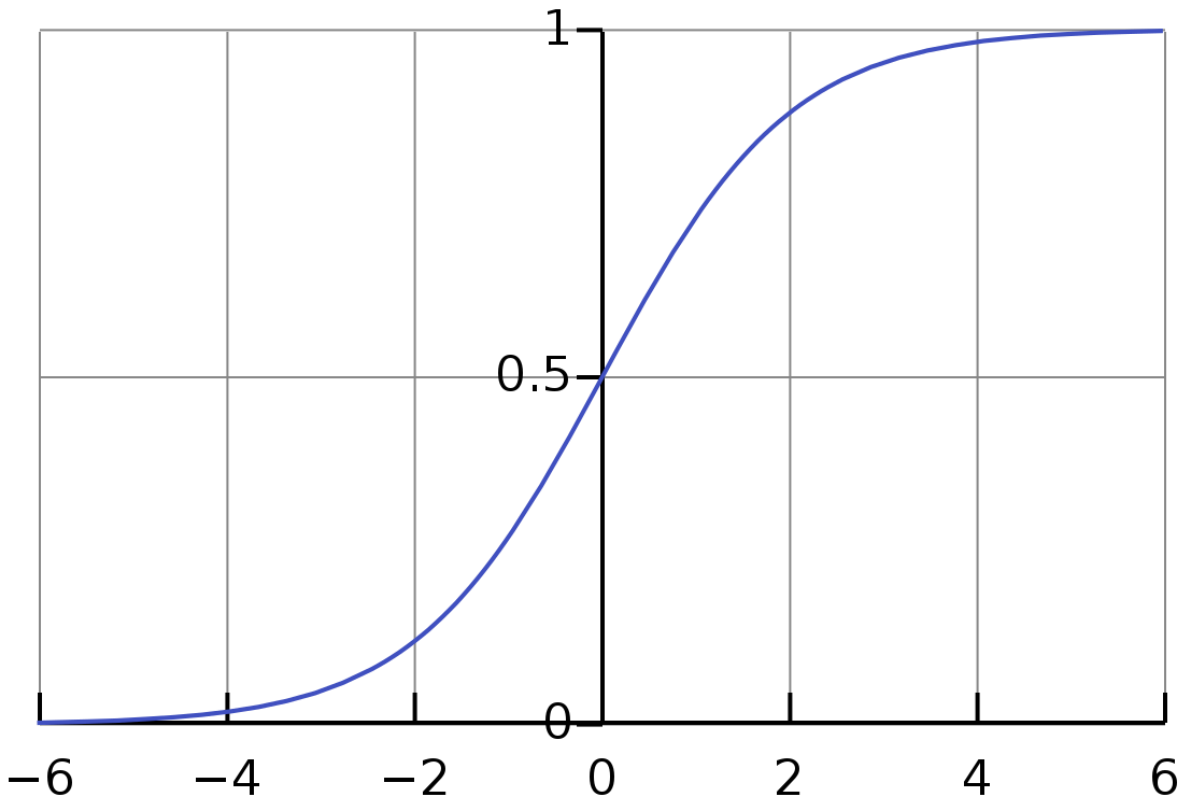
A neural network with a linear activation function is simply a linear regression model. It has limited power and ability to handle complexity varying parameters of input data.

Nonlinear Activation Functions

1. They allow backpropagation because they have a derivative function which is related to the inputs.

2. They allow "stacking" of multiple layers of neurons to create a deep neural network. Multiple hidden layers of neurons are needed to learn complex data sets with high levels of accuracy.

Sigmoid/Logistic Function



Sigmoid/Logistic Function

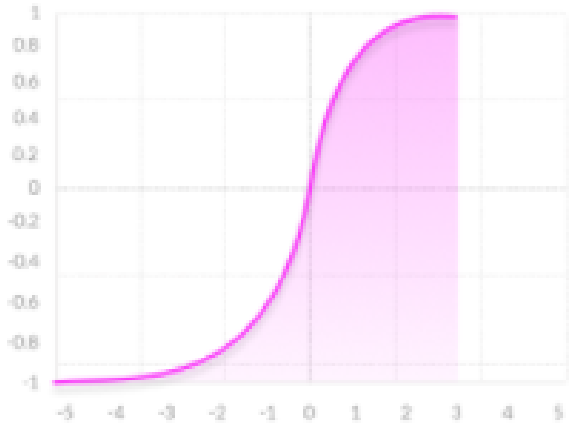
Advantages

- **Smooth gradient**, preventing "jumps" in output values.
- **Output values bound** between 0 and 1, normalizing the output of each neuron.
- **Clear predictions**—For X above 2 or below -2, tends to bring the Y value (the prediction) to the edge of the curve, very close to 1 or 0. This enables clear predictions.

Disadvantages

- **Vanishing gradient**—for very high or very low values of X, there is almost no change to the prediction, causing a vanishing gradient problem. This can result in the network refusing to learn further, or being too slow to reach an accurate prediction.
- **Outputs not zero centered**.
- **Computationally expensive**

TanH / Hyperbolic Tangent



Advantages

- **Zero centered**—making it easier to model inputs that have strongly negative, neutral, and strongly positive values.
- Otherwise like the Sigmoid function.

Disadvantages

- Like the Sigmoid function

ReLU (Rectified Linear Unit)



Advantages

- **Computationally efficient**—allows the network to converge very quickly
- **Non-linear—**although it looks like a linear function, ReLU has a derivative function and allows for backpropagation

Disadvantages

- **The Dying ReLU problem**—when inputs approach zero, or are negative, the gradient of the function becomes zero, the network cannot perform backpropagation and cannot learn.

THANK YOU